Arista Access Node Optical Transport Failure Resolution Procedure for WFD-ACC-71 at Whitefield POP

Node: WFD-ACC-71 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3397

Description

The Optical Transport Failure (ALARM_3397) in the access role of the Arista node WFD-ACC-71 at Whitefield POP is precipitated by a concatenation of factors intrinsic to the optical transport layer. This failure domain is primarily occasioned by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The Arista access node, configured to operate in conjunction with the broader optical transport network, relies on the seamless interaction between the physical layer (Layer 1) and the data link layer (Layer 2) of the OSI model. The failure of the optical transport can be attributed to the inherent susceptibility of optical fibers to physical damage, the degradation of signal strength over distance, or the malfunction of the SFP/XFP modules which are critical for converting electrical signals to optical signals and vice versa. The Arista EOS (Extensible Operating System) provides a robust platform for diagnosing and resolving such failures through its comprehensive set of CLI commands. Understanding the intricacies of the optical transport layer and the specific roles of components such as SFP/XFP modules, optical fibers, and the Arista node itself is crucial for effectively addressing and resolving the Optical Transport Failure. The resolution of this failure requires meticulous adherence to a structured procedure that involves preliminary checks, diagnostic tests, and targeted interventions, all of which must be executed within the context of the Arista CLI dialect to ensure compatibility and efficacy.

Pre-requisite Checks

- Dispatch remote hands to Whitefield POP for physical inspection and potential interventions.

- Verify the status of redundant links to ensure network resilience.

- Confirm the accessibility of necessary tools and replacement parts (SFP/XFP modules, fiber optic cables).

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

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Command: show interfaces transceiver detail, Expected Output: Detailed information about the SFP/XFP modules, including their status and
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signal strength., Description: This command provides insight into the health and performance of the SFP/XFP modules, which can indicate degradation or failure. Analyzing the output helps in isolating issues related to the optical transport layer.

Command: show interfaces ethernet counters, Expected Output: Interface-specific counters, including errors and discards, which can indicate signal integrity issues., Description: By examining the interface counters, network administrators can identify potential issues with the optical signal, such as high error rates or discards, which could point to problems with the fiber or the SFP/XFP modules.
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Resolution Steps

Step 1: Step: Replace Faulty SFP/XFP Module, Command: replace module with , Description: If the diagnostic process identifies a faulty SFP/XFP module, replacing it with a functional one is necessary. This step involves physically removing the defective module and inserting a new one, followed by the execution of the replace command to update the system configuration.

Step 2: Step: Adjust Rx/Tx Power Levels, Command: interface ethernet transceiver power-level , Description: In cases where the issue is related to Rx/Tx power loss, adjusting the power levels of the SFP/XFP module can help stabilize the optical signal. The specific power level adjustment depends on the module type and the optical link characteristics.

Post-Check Verification

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Command: show interfaces transceiver detail, Expected Output: Updated information reflecting the replacement or adjustment made, with improved signal strength or module status., Description: Re-executing the transceiver detail command after the replacement or adjustment helps verify that the issue has been resolved and the optical transport layer is functioning as expected.

Command: show interfaces ethernet counters, Expected Output: Reduced or eliminated error counters, indicating improved signal integrity., Description: Checking the interface counters again after the resolution steps confirms whether the adjustments have positively impacted the signal quality and reduced errors.
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Roll-back Procedures

- Step: Revert to Original SFP/XFP Module, Command: replace module with , Description: If the replacement module does not resolve the issue or introduces new problems, reverting to the original module may be necessary. This step involves the physical removal of the new module and the reinsertion of the original one, followed by the execution of the replace command.

- Step: Reset Rx/Tx Power Levels to Default, Command: default interface ethernet transceiver power-level, Description: If adjusting the power levels does not resolve the issue, resetting them to their default values can help revert the configuration to a known good state. This command removes any custom power level settings, returning the module to its factory default configuration.